



Data Preparation for Business Analytics

A course with R

*2026 (fresh) edition

Balázs KOTOSZ

balazs.kotosz@ext.neoma-bs.fr

Instructor info



- **Education and academic experience:**
 - BBA, University of Miskolc, Hungary
 - MSc Financial Management, ESC Rouen (today **NEOMA**), France
 - Ecole Polytechnique (engineering cycle), France
 - Ph.D. in Economics, Corvinus University, Budapest, Hungary
 - Machine Learning, Stanford, with Andrew Ng
 - Actually:
 - Freelance Data Scientist
 - Teaching data analysis

Course content

- Starting with R, R Studio, data types and structures, packages
- Importing data
- Cleaning data:
 - Problems with data types, duplicates, dates
 - Categorical and text data
 - Numerical data, missing values, outliers
 - Formatting datasets: subsets, merging, nesting
- Complex case study

Assessment

- 50% continuous assessment:
 - 30% daily exercises (quizzes and code submission)
 - 5 graded MCQ on paper, no computer, no phone. Paper-based notes can be used (dates not announced)
 - At the end of each class, you will be asked to upload your classwork: your notes and the exercise solutions
 - Show your personal contribution, your code, your comments for yourself and your ideas to solve the exercises
 - 20% last day project
 - A database to clean without explicit instructions
- 50% final exam:
 - On paper, problem solving
 - Problem detection, short code writing, interpretation
 - An eight-page long (four A4-size sheets), printed cheat sheet is allowed (it is your responsibility to edit the document and bring along)
- All assessments are individually evaluated, except the project (groups of 3 maximum).

Last day project instructions

- Transform the attached data file to an .R data frame and clean it to be ready for analysis. Keep in mind tidy data rules and use the tricks you learnt in the course.
- You may find problems with the structure of the data, data types, and missing values, outliers, wrong numbers, spelling problems or other issues. Try to detect all and solve them. Don't forget to explain what you found, why you chose a solution (e.g. for missing values, you can delete observations/variables, you can ignore them, you can impute them using different methods).
- You will have access to personal help during the class time, but you can work on the project already before the beginning of the class.
- Submit here your project code in the form of an .R script. No other formats are accepted
- **You can work in groups of 3 maximum.** Working alone is not recommended, but also acceptable. One submission per team is required. In the first comment, list the team members with their names (as in the official records of the school). If you shared the work, and some parts are the work of one student, you can also mention this fact in a comment.
- If you used Generative AI to solve the problem, you should add the prompts and the answers either at the end of the script (in comments), or in a separate pdf file.

Accepted formats

- For any submission where a script is needed, only:
 - .R (preferred)
 - .Rmd*files are accepted
- In some rare cases, data file submission may be required. Pay attention to the required format, it may change from exercise to exercise.
- Any file submitted in a wrong format will NOT be evaluated and its score will be 0. Screenshots, images of the code, a code copy-pasted to Word or pdf files will be automatically rejected.
- Code in other programming languages is not accepted even if it works (see next slide).

* Rmd is for Rmarkdown. If you are familiar with it, you can use it, on your own responsibility.

We need R in RStudio

0. Be sure that you have at least 20GB free storage space on your hard disk and you have a PC or a Mac.

- Tablets, Ipads, Chromebooks are NOT welcome, you risk not to be able to work

1. Install R (the version that fits your operating system):

- <https://cran.r-project.org/>
- Choose the version adapted to your operating system (Linux/MacOS/Windows) and follow the link
- Download the base (for Windows)
- Download the .pkg file that is adapted to your Mac (you can check the chip version by clicking on the Apple symbol on your Mac)
- When R is installed ->

2. Install RStudio (the free version):

- <https://posit.co/download/rstudio-desktop/>
- Normally, it will recognize your operating system, you can still scroll down and find the right version
- Don't pay for it, choose the free version

3. If R and/or RStudio has already been installed on your computer, check if it is the latest version

- If not, go to 1
- If yes, update also your packages
- If you have a version with unsolved conflicts, start again with a clean install